

Setup

The following code sets up the system by downloading the library from the web.

Make a closed figure function

Here is a list of colors: "blue", "green", "red", "cyan", "magenta", "yellow", "black", "white", "olive", "indigo".

Make a video of drawing a face

```
In [7]: b=4
        d=5

        plot = graph_funs(equal_axes=True, hide_axes=False)

        plot.set_XY(minX=-10, maxX=+10, minY=0, maxY=+10)

        make_closed_fig(plot, a=6, b=1.1, h=0, k=-5, d=d, num_of_points=20, color="brown")
        make_closed_fig(plot, a=6, b=1.1, h=0, k=-5, d=3, num_of_points=20, color="tan")
        plot.plot_funs(plot_title="face", img_name="frame1.png")

        make_closed_fig(plot, a=1, b=2, h=2.7, k=5.5, d=1, num_of_points=20, color="brown")
        make_closed_fig(plot, a=1, b=2, h=-2.7, k=5.5, d=1, num_of_points=20, color="brown")
        plot.plot_funs(plot_title="ears", img_name="frame2.png")

        make_closed_fig(plot, a=1, b=2, h=0, k=0.5, d=1, num_of_points=20, color="#ff5aa4")
        plot.plot_funs(plot_title="big nose", img_name="frame3.png")

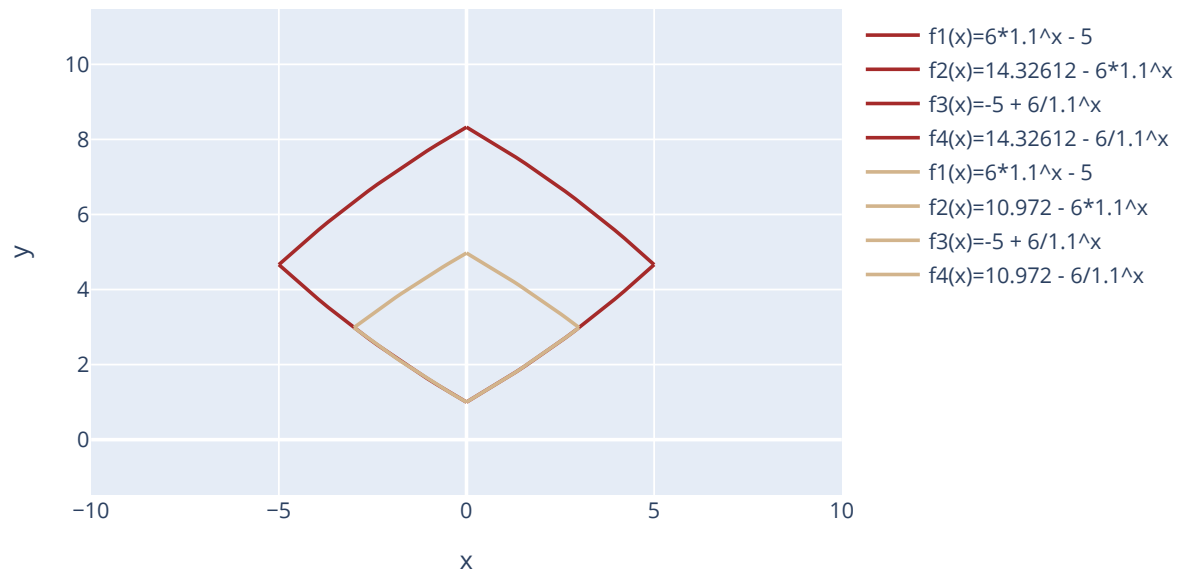
        make_closed_fig(plot, a=1, b=2, h=-2.7, k=5.5, d=.5, num_of_points=20, color="#FF0080")
        make_closed_fig(plot, a=1, b=2, h=2.7, k=5.5, d=.5, num_of_points=20, color="#FF0080")
        plot.plot_funs(plot_title="small ears", img_name="frame4.png")

        make_closed_fig(plot, a=1, b=2, h=2.5, k=3, d=.8, num_of_points=20, color='black')
        make_closed_fig(plot, a=1, b=2, h=-2.5, k=3, d=.8, num_of_points=20, color='black')

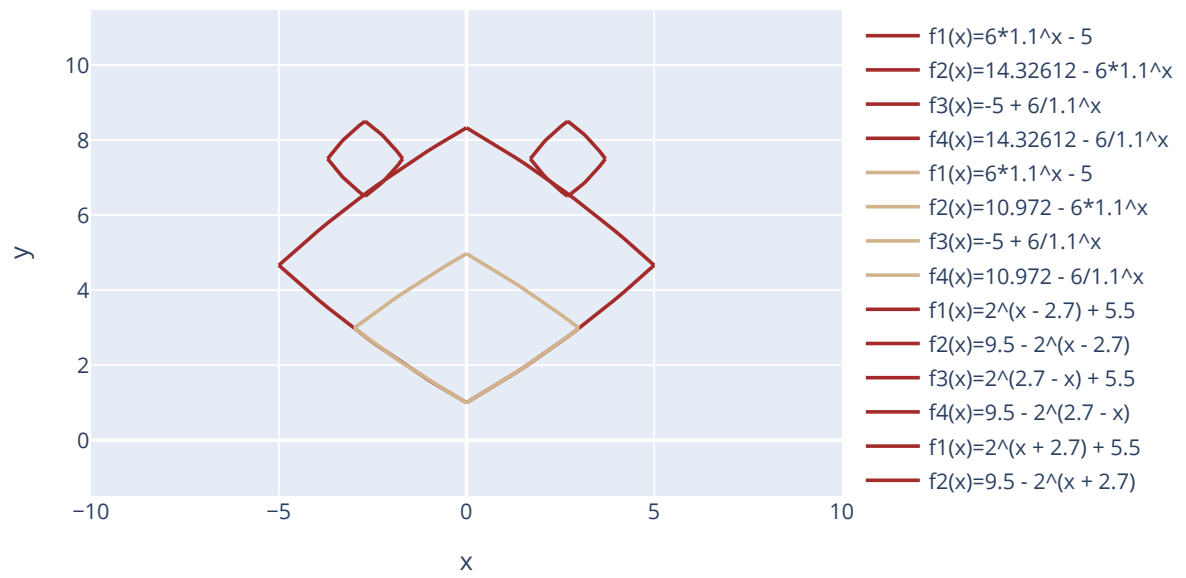
        make_closed_fig(plot, a=1, b=2, h=2.5, k=3, d=.4, num_of_points=20, color="#06402b")
        make_closed_fig(plot, a=1, b=2, h=-2.5, k=3, d=.4, num_of_points=20, color="#06402b")

        plot.plot_funs(plot_title="eyes", img_name="frame5.png")
```

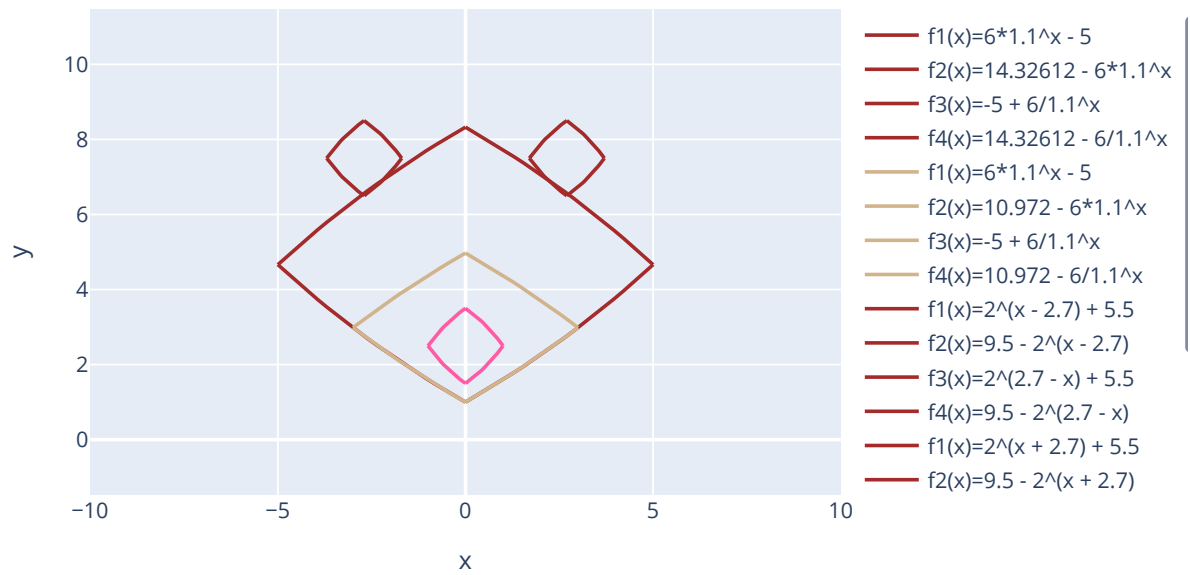
face



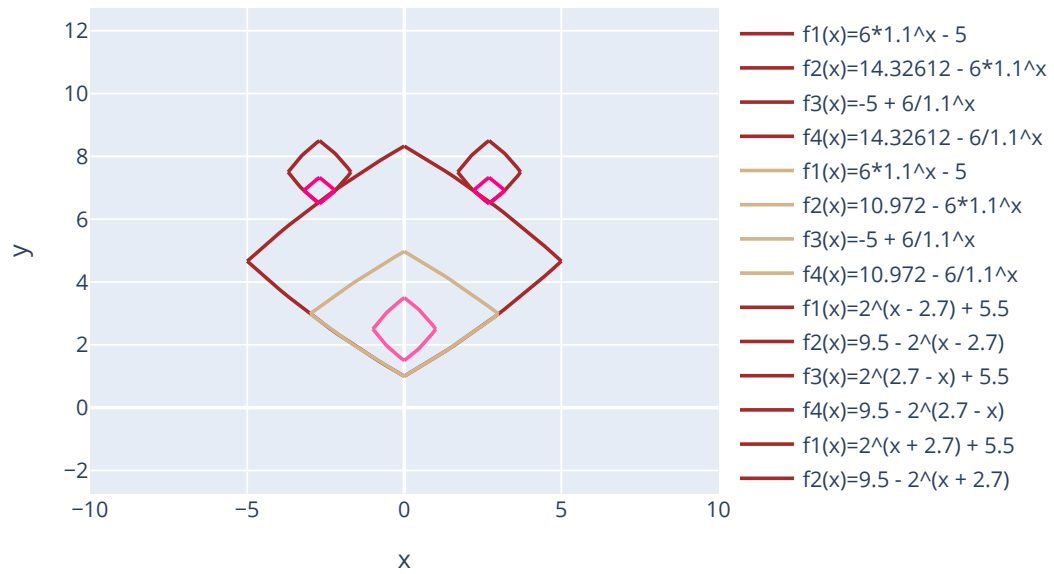
ears



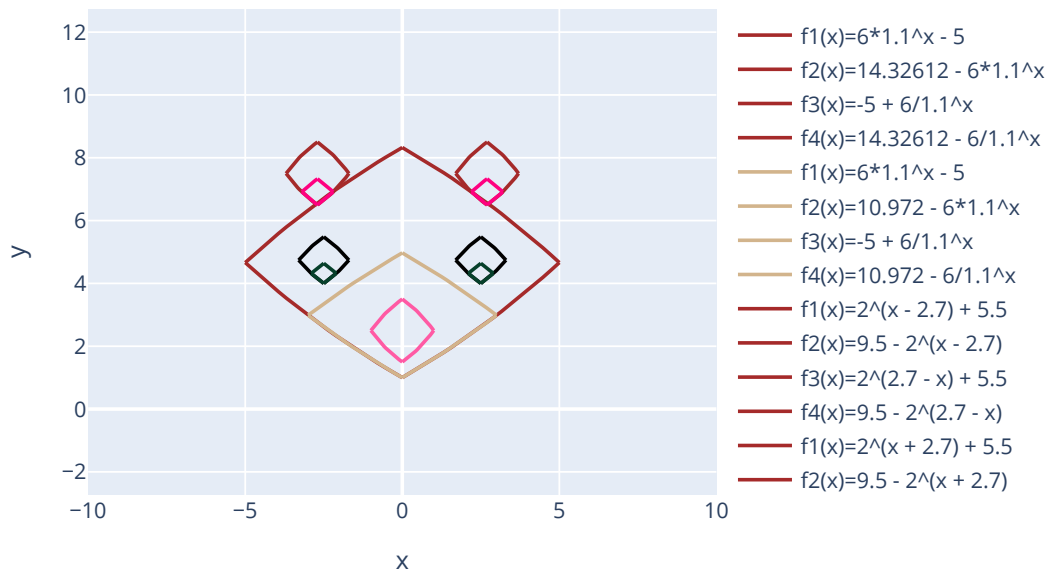
big nose



small ears



eyes



In [8]: `!jupyter nbconvert --to html /content/drive/MyDrive/Colab_Notebooks/Untitled0.ipynb`

[NbConvertApp] Converting notebook /content/drive/MyDrive/Colab_Notebooks/Untitled0.ipynb to html

[NbConvertApp] Writing 418429 bytes to /content/drive/MyDrive/Colab_Notebooks/Untitled0.html

In []: